

Breeze-TTS-Production Progress

Date : 2026/5/8

Reference Audio Sources

The number of speakers must be balanced.

- **Too few speakers:** the data distribution is not diverse enough.
- **Too many speakers:** each speaker only has a few seconds of audio, which makes it hard for the model to learn a good speaker embedding.

To manage all speakers, we maintain a `speaker_table` with the following fields:

Field Name	Example
row_id	1
speaker_id	00335f44eef95697c18b8674efb2e701745e95221777216ecb5996673ac4e10c875b6e078dbaf462dd9104cbc3b57
wavFile	common_voice_zh-TW_31049337
transcript	這時期應該是個人套餐
wav_length	2.1 sec
source	CommonVoice25
total_sentences	0
total_audio_seconds	0
created_date	2026-04-16
gender	male_masculine
age	thirties
speech_rate	TBD

The current audio sources are listed below.

Source	# Speakers	Filtering Conditions
MediaTek-Research/formosaspeech	417	postprocess = “無混雜訊” duration_sec < 20 text not empty audio-text pairs are consistent
Common Voice Scripted Speech 25.0 - Chinese (Taiwan)	2241	TBD

Reference Text Sources

Control the number of unique sentences to ensure balanced coverage of Chinese characters and Bopomofo as much as possible.

Sentence duration should also follow a reasonable distribution between 1–20 seconds.

To manage all sentences, we maintain a `sentence_table` with the following fields:

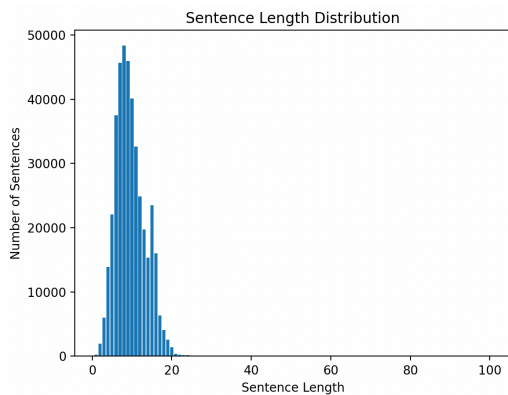
Field Name	Example
row_id	28687
sentence_id	0dc7b2e1174642385c81554762f586df4c8b34e538feb31bdcdf57659a6c6c4b
sentence	有某種程度是對的
source_type	YouTube Subtitles
source_name	boyin-EP201
sentence_length	8
used_count	0

Field Name	Example
created_date	2026-04-16

The current audio sources are listed below. There are a total of 409,949 unique sentences.

Type	Source	# Sentences	Ratio
YouTube Subtitles	只能喝酒的圖書館	72,000	17.56%
YouTube Subtitles	志祺七七	72,000	17.56%
YouTube Subtitles	博音	61,905	15.10%
YouTube Subtitles	Sherry's Notes 雪力的心理學筆記	58,051	14.16%
YouTube Subtitles	范琪斐的美國時間	51,368	12.53%
YouTube Subtitles	唐綺陽官方專屬頻道	45,514	11.10%
Dataset	CommonVoice25	20,527	5.01%
YouTube Subtitles	Cofit x 宋晏仁醫師 - 初日醫學	16,245	3.96%
YouTube Subtitles	High A Day	11,093	2.71%
Dataset	MediaTek-Research/formosaspeech	1,246	0.30%

The distribution of sentence lengths is shown in the figure below.



Audio Sources & Text Sources Pairing

Each speaker samples 3,500 sentences from the `sentence_table`, resulting in an expected contribution of approximately 3.8 hours of speech per speaker. The sampling process follows a predefined sentence-length distribution, where different numbers of sentences are drawn from each length bucket to balance short and long utterances.

The detailed sampling rules are shown in the table below.

Sentence Length	Ratio (%)	# Sentences	# Sentences per Speaker	Allocation Ratio(%)	#Repeats per Sentence	Expected Size
1-4	4.59	18,805	175	5.00%	24.74	465,150
5-8	35.83	146,882	1,200	34.29%	21.72	3,189,600
9-14	42.98	176,209	1,375	39.29%	20.74	3,654,750
15-20	15.57	63,818	700	20.00%	29.15	1,860,600
21-30	0.86	3,519	40	1.14%	30.21	106,320
31-45	0.09	363	4	0.11%	29.29	10,632
46-60	0.03	104	2	0.06%	51.12	5,316
61-80	0.03	114	2	0.06%	46.63	5,316
80+	0.03	135	2	0.06%	39.38	5,316
Total	100	409,949	3,500	100.00%		9,303,000

To manage all pairing data, we maintain a `pairing_table` with the following fields:

Field Name	Example
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speaker_prompt_audio_filename	common_voice_zh-TW_21270221
speaker_id	816aa1e278293a2b39d799a4046b832b64c6c8bf92786ac149cde1ff8df3668521f047ea3678cf5c2f
speaker_prompt_text_transcription	想把所有能收到的發票都存入載具內
content_to_synthesize	受到影響的人
sentence_id	83c0b39bc3d24640e567882b841f4298335fb0da4aa6cb89de8f33762c98223a
output_audio_filename	SD0003938210

Training Data Filtering

ASR

This code uses the [faster-whisper](#) model to perform speech-to-text transcription on `.wav` audio files. The example output includes the detected language, language confidence probability, audio duration, and transcribed text.

We retain samples with a Character Error Rate (CER) below 5%. In experiments conducted on 114,820 `.wav` files, the retention rate was approximately 82.82%.

Speaker Embedding Similarity

Using [3D-Speaker](#), we compute the embedding similarity between the reference speaker audio and the generated audio, with scores ranging from 0 to 1. We then apply an appropriate threshold to retain samples with higher speaker similarity.